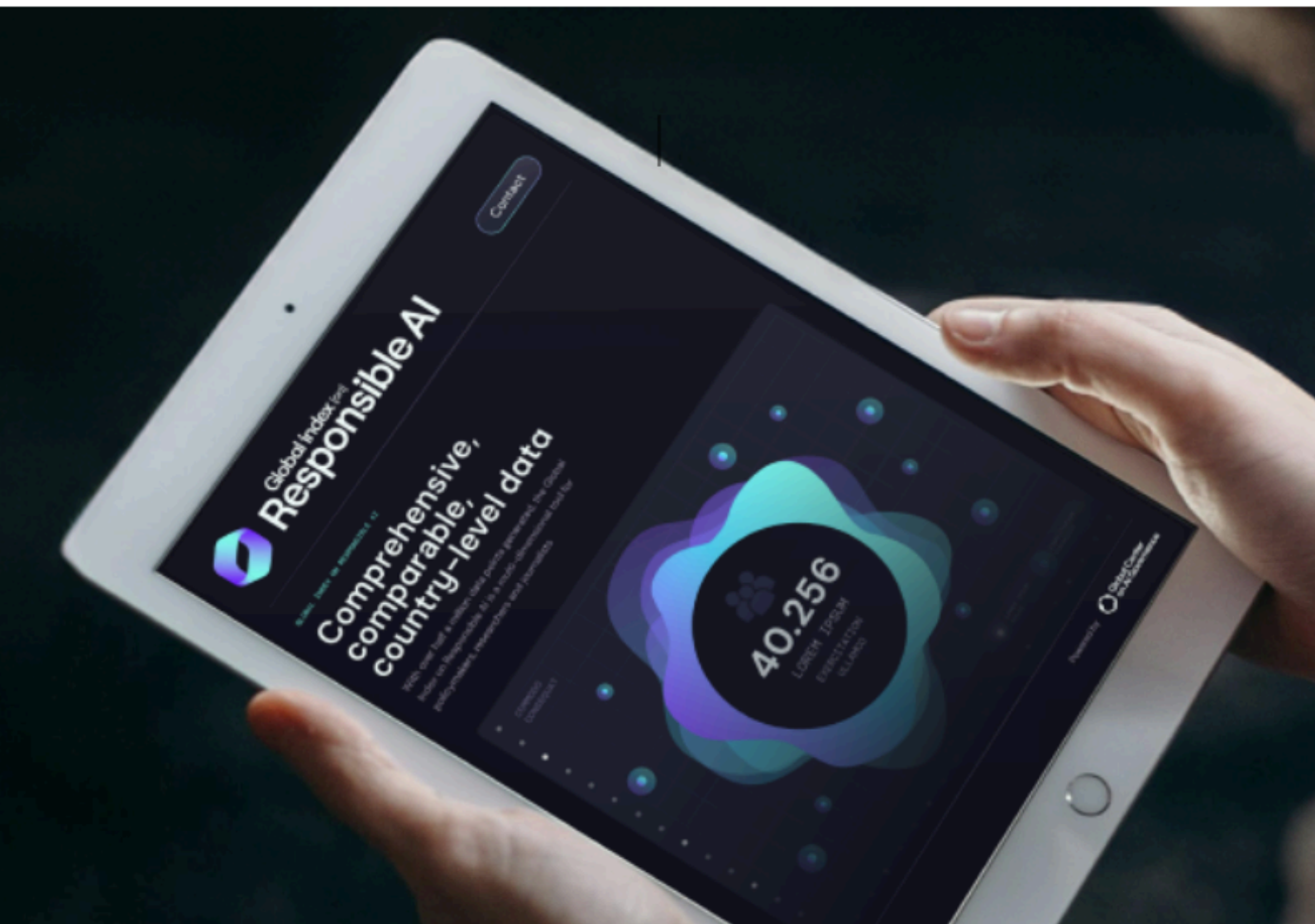




# Methodology and Conceptual Framework of the Global Index on Responsible AI.

1st Edition

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## **Building the Global Index on Responsible AI**

The structure and content of the Global Index on Responsible AI (GIRAI) emanates from a series of consultations with a wide range of stakeholders from around the world, specifically those from Africa, Asia, the Caribbean and South and Central America. During these consultations, participants were invited to engage with the scope and structure of the GIRAI, which not only included the conceptual foundation and rationale for the Index, but also areas of measurement, including the three pillars, three dimensions and thematic areas. Participants were invited to comment on the suggested areas of measurement and the interpretation thereof, as well as the approaches to data collection, analysis, and reporting. Specific attention was paid to relevance of each thematic area across countries, as well as the local prioritisation of different human rights groups and the potential availability of data in the country.

In September 2023, a pilot of the GIRAI was completed in 10 countries, including Costa Rica, Kenya, Jamaica, Canada, Serbia, India, Sri Lanka, Palestine, Burkina Faso, and Georgia. Preliminary findings from the pilot study validated the Global Index's methodology and demonstrated its proof of concept.

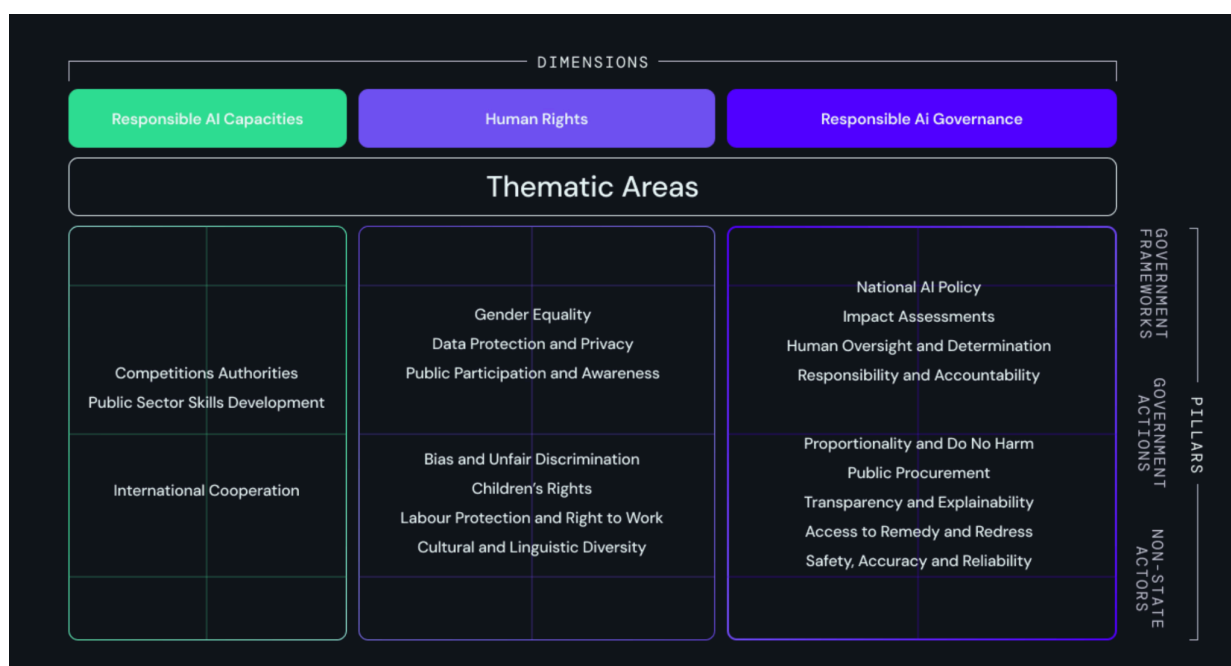
The countries included in the 1st Edition of the Global Index on Responsible AI were chosen based on where suitable country-researchers were recruited and the existing research network of institutional partners. Governments were not involved in determining whether their country was included in the Index.

Data was collected for Chinese Taipei (commonly referred to as Taiwan) and the West Bank, which is under the administration of the Palestinian Authority. No data collection took place in Gaza. We recognize that these are, at the time of writing, disputed jurisdictions.

## Index structure

The Global Index on Responsible AI is composed of:

- 3 dimensions that align with human rights-based standards and democratic principles,
- 3 pillars accounting for the AI ecosystem,
- 19 thematic areas covering core components of the concept of Responsible AI and,
- 57 indicators measuring the performance of each thematic area within each pillar.



## Dimensions

The Global Index on Responsible AI is composed of three Dimensions rooted in human rights and democratic principles, these are:

### Responsible AI Governance

This dimension measures the degree to which country-level governance regimes uphold effective and rights-preserving practices in responsible AI.

## **Human Rights & AI**

The protection of and respect for fundamental human and collective rights and freedoms is paramount to the responsible development and use of AI around the world. This dimension measures the extent to which countries are taking steps to protect, promote and respect key human rights implicated by AI.

## **Responsible AI Capacities**

In order to meaningfully advance on responsible AI commitments and policy agendas governments require certain capacities. This dimension measures the extent to which the key state capacities required to advance responsible AI exist and are being met and promoted.

## **Thematic Areas**

Each dimension of the GIRAI is divided into Thematic Areas that cover the core components of responsible AI. Thematic areas were developed following a deep dive review of literature on responsible AI and consultations with stakeholders working on AI and human rights across the world, and particularly from the Global South, where their views were less represented in published literature. The UNESCO Recommendation on the Ethics of AI and the OECD Principles on AI were the key foundational documents relied on in determining the thematic areas for the GIRAI.

Thematic Areas are grouped below under their respective dimension. Definitions of each can be accessed by clicking on the respective name.

### **Responsible AI Governance**

1. [National AI Policy](#)
2. [Impact Assessments](#)
3. [Human Oversight and Determination](#)
4. [Responsibility and Accountability](#)
5. [Proportionality and Do No Harm](#)
6. [Public Procurement](#)
7. [Transparency and Explainability](#)
8. [Access to Remedy and Redress](#)
9. [Safety, Accuracy and Reliability](#)

### **Human Rights & AI**

10. [Gender Equality](#)
11. [Data Protection and Privacy](#)

12. [Bias and Unfair Discrimination](#)
13. [Children's Rights](#)
14. [Labour Protection and Right to Work](#)
15. [Cultural and Linguistic Diversity](#)

#### Responsible AI Capacities

16. [Competitions Authorities](#)
17. [Public Sector Skills Development](#)
18. [International Cooperation](#)

## Pillars

Each thematic area was assessed through three pillars of evidence, which account for the AI ecosystem in relation to the thematic area being assessed. These pillars are:

### Government frameworks

Government frameworks are defined to include the laws, regulations, policies, strategies and/or guidelines adopted by the national or federal government that address the implications of AI with respect to a particular thematic area.

### Government actions

This pillar focuses on government actions that involve either framework development, the implementation of frameworks, or government initiatives to protect or promote the rights or principles identified in the thematic area even in the absence of a government framework.

### Non-state actors

This pillar assesses actors outside of government (including universities, civil society organisations, and private sector entities) working on issues related to AI and the thematic area.

## Indicators

Thematic areas are measured across the three pillars: Government Frameworks, Government Actions and Non-state Actors. Consequently, each Thematic Area is associated with three standard indicators.

The table below summarises the dimensions, thematic areas, pillars and indicators used in the 1st edition of the Global Index on Responsible AI.

|                           |                                   | PILLARS                                                   |                                                        |                                                      |
|---------------------------|-----------------------------------|-----------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------|
|                           |                                   | Government Frameworks                                     | Government Actions                                     | Non-state Actors                                     |
| DIMENSIONS                | THEMATIC AREAS                    | INDICATORS                                                |                                                        |                                                      |
| Responsible AI Governance | National AI Policy                | National AI Policy – Government frameworks                | National AI Policy – Government actions                | National AI Policy – Non-state actors                |
|                           | Impact Assessments                | Impact Assessments – Government frameworks                | Impact Assessments – Government actions                | Impact Assessments – Non-state actors                |
|                           | Human Oversight and Determination | Human Oversight and Determination – Government frameworks | Human Oversight and Determination – Government actions | Human Oversight and Determination – Non-state actors |
|                           | Responsibility and Accountability | Responsibility and Accountability – Government frameworks | Responsibility and Accountability – Government actions | Responsibility and Accountability – Non-state actors |

|                     |                                    | Government Frameworks                                      | Government Actions                                      | Non-state Actors                                      |
|---------------------|------------------------------------|------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------|
| DIMENSIONS          | THEMATIC AREAS                     | INDICATORS                                                 |                                                         |                                                       |
|                     | Proportionality and Do No Harm     | Proportionality and Do No Harm – Government frameworks     | Proportionality and Do No Harm – Government actions     | Proportionality and Do No Harm – Non-state actors     |
|                     | Public Procurement                 | Public Procurement – Government frameworks                 | Public Procurement – Government actions                 | Public Procurement – Non-state actors                 |
|                     | Transparency and Explainability    | Transparency and Explainability – Government frameworks    | Transparency and Explainability – Government actions    | Transparency and Explainability – Non-state actors    |
|                     | Access to Remedy and Redress       | Access to Remedy and Redress – Government frameworks       | Access to Remedy and Redress – Government actions       | Access to Remedy and Redress – Non-state actors       |
|                     | Safety Accuracy and Reliability    | Safety Accuracy and Reliability – Government frameworks    | Safety Accuracy and Reliability – Government actions    | Safety Accuracy and Reliability – Non-state actors    |
| Human Rights and AI | Gender Equality                    | Gender Equality – Government frameworks                    | Gender Equality – Government actions                    | Gender Equality – Non-state actors                    |
|                     | Data Protection and Privacy        | Data Protection and Privacy – Government frameworks        | Data Protection and Privacy – Government actions        | Data Protection and Privacy – Non-state actors        |
|                     | Public Participation and Awareness | Public Participation and Awareness – Government frameworks | Public Participation and Awareness – Government actions | Public Participation and Awareness – Non-state actors |



|                           |                                     | Government Frameworks                                       | Government Actions                                       | Non-state Actors                                       |
|---------------------------|-------------------------------------|-------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------|
| DIMENSIONS                | THEMATIC AREAS                      | INDICATORS                                                  |                                                          |                                                        |
|                           | Bias and Unfair Discrimination      | Bias and Unfair Discrimination – Government frameworks      | Bias and Unfair Discrimination – Government actions      | Bias and Unfair Discrimination – Non-state actors      |
|                           | Children's Rights                   | Children's Rights – Government frameworks                   | Children's Rights – Government actions                   | Children's Rights – Non-state actors                   |
|                           | Labour Protection and Right to Work | Labour Protection and Right to Work – Government frameworks | Labour Protection and Right to Work – Government actions | Labour Protection and Right to Work – Non-state actors |
|                           | Cultural and Linguistic Diversity   | Cultural and Linguistic Diversity – Government frameworks   | Cultural and Linguistic Diversity – Government actions   | Cultural and Linguistic Diversity – Non-state actors   |
| Responsible AI Capacities | Competitions Authorities            | Competitions Authorities – Government frameworks            | Competitions Authorities – Government actions            | Competitions Authorities – Non-state actors            |
|                           | Public Sector Skills Development    | Public Sector Skills Development – Government frameworks    | Public Sector Skills Development – Government actions    | Public Sector Skills Development – Non-state actors    |
|                           | International Cooperation           | International Cooperation – Government frameworks           | International Cooperation – Government actions           | International Cooperation – Non-state actors           |

Data for each Thematic Area was collected from desk-research of policy documents, government reports and expert key-informant interviews conducted by country-level researchers, representing what is referred to as primary data.

Primary data has been preferred over secondary data for measuring country-level progress due to its specificity, relevance, and reliability to the research question at hand. Currently there is no index measuring first-hand the commitments and practices of governments on responsible AI principles, but instead measure technical AI capacity and performance. The collection of primary data filled major data gaps on responsible AI commitments and practices, especially across LMICs. Further, the establishment of a global network of responsible AI experts function as change agents in the countries in which they are working, utilising the information and data gathered from the GIRAI to advocate for government action and stronger rights-protections. Since their appointment with the GIRAI, many of the country-experts have been called on by their governments to provide expert input into AI policy processes.

## **Pillar Coefficients**

While primary data captures and assesses the presence of evidence within each Pillar and Dimension at the Thematic Area level, in order to account for the effectiveness of Government Frameworks, Government Actions, and the enabling environment for Non-state Actors in each country, a specific coefficient was designed using secondary data. Secondary data was added as contextual information to assess country-level capacity and historical performance related to the realisation and impact of government commitments. This contextual layer supplements the primary data with insights into past government performance in key areas related to each pillar, including fundamental human rights tracking records.

The Government Frameworks and Government Actions coefficients were computed using data from the [Worldwide Governance Indicators](#) produced by the World Bank, while the Non-State Actors coefficient was developed using metrics from [Freedom in the World Index](#), built by the Freedom House. These coefficients were built from secondary data sources as the primary data collection was focused solely on gathering evidence directly related to responsible AI, and where relevant and valid secondary data existed, it was not necessary to collect primary data. The coefficients also allowed for the study to include assessment of the potential human-rights related impact of AI at a country-level, where it is currently not possible to assess and measure.

The specific series used to build each coefficient are detailed below:

- Frameworks coefficient: [Regulatory Quality](#) and [Rule of Law](#).
- Government actions coefficient: [Government Effectiveness](#) and [Control of Corruption](#).
- Non-state actors coefficient: [Freedom of Expression](#) and [Belief and Associational & Organisational Rights](#).

### Missing values

For countries with some missing data for the coefficient calculation, the peer group mean value, defined as their geographical subregion, was used to impute the missing data.

## Scores Calculation

The calculation of the GIRAI score involves a multi-step process. Within each pillar, indicator's values are calculated by translating qualitative survey data into numerical values, combining them into indicators which are then normalised to a common scale (0-100) and integrated with the coefficients. Finally, the GIRAI score is derived as the weighted average of the pillar scores.

Furthermore, since each indicator is assigned to one pillar and one dimension, it enables the computation of scores for pillars, dimensions, and thematic areas.

### Step 1: Calculating the Indicators

The first step involves calculating the indicators within each thematic area at the level of each pillar.

#### *Government Frameworks Indicators*

The variables used for this pillar are “existence” and “type of framework”, with the type of framework serving as a proxy for the level of enforceability. Thus, frameworks with a higher level of enforceability are given a higher score, such as:

- 3 points: legally enforceable frameworks such as laws, by-laws, decrees, resolutions and other types of regulations.

- 2 points: policies, strategies, long term strategic plans, etc. that articulate the approach adopted by the government and are expected to be accomplished.
- 1 point: non-binding frameworks that provide guidelines, principles, recommendations, which are voluntary measures that implementers should follow, but are not required to.
- 0 points: no framework addressing the thematic area.

#### *Government Actions Indicators*

The variables used for this indicator are “existence” and “type of action”, with the type of action scored according to the following rules:

- 1 point: if the country has an adopted framework in place for the thematic area, the government action in relation to that thematic area is taken as a proxy of implementation of an existing framework by the state.
- 1 point: if the country does not have an adopted framework in place for the thematic area, but the government action is in relation to developing one, the action is seen as a proxy of policy development by the state
- 0.9 points: cases where the country already has a framework in place and the action identified is developing a new framework; or where a framework was not identified but still there’s evidence of government actions different from developing a new framework.
- 0 points: no evidence of government actions found in respect of the thematic area.

#### *Non-state Actors Indicators*

The variables used to score this indicator is the “existence” and “type of actors”, with the type of actor (academia, civil society and private sector) serving as proxies for diversity. Because each of these actors is likely to represent a different cohort of interests (e.g. the private sector may be interested in responsible AI R&D, while civil society organisations may be more interested in protecting people’s right to privacy), countries that have a more diverse group of non-state actors conducting responsible AI initiatives scored higher.

- 3 points: presence of all three types of actors.
- 2 points: presence of at least two types of actors.

- 1 point: presence of at least one type of actor.
- 0 points: no involvement from non-state actors.

### **Step 2: Normalising the Values**

The second step involves bringing the values to a common range. This is achieved by using a minimum-maximum normalisation formula: the indicator score obtained in the previous step is normalised by subtracting from this “raw” value the theoretical minimum score (0), and dividing the result by the difference between the theoretical maximum and the theoretical minimum value for the indicator. Finally, this result is multiplied by 100.

Below are detailed the minimum and maximum theoretical value for each type of indicator:

- Government actions indicators range from 0 to 1.
- Non-state actors indicators range from 0 to 3.
- Frameworks indicators range from 0 to 3, but the maximum for rescaling depends on the thematic area. This is because different degrees of enforceability are necessary to protect the right or principle evaluated in the thematic area.
  - Therefore, for thematic areas where legally-enforceable frameworks are the benchmark, the theoretical value is set to 3 points.
    - These thematic areas are: Access to Remedy and Redress, Bias and Unfair Discrimination, Children's Rights, Competitions Authorities, Data Protection and Privacy, Gender Equality, Labour Protection and Right to Work, Proportionality and Do No Harm, Public Participation and Awareness and Public Procurement.
  - For the rest of the thematic areas, the theoretical maximum is set to 2 points.

Once indicators are normalised, their values are adjusted by the corresponding coefficients in the next step.

### **Step 3: Adjusting Normalised Indicators Values by the Coefficients**

The third step involves adjusting the normalised values with the country-specific coefficient for each pillar. Coefficients range from 0 – 1, and are obtained as follows:

#### *Frameworks Coefficient*

For each country, it is computed as the arithmetic average of the values from the Regulatory Quality and Rule of Law metrics divided by 100.

#### *Government Actions Coefficient*

For each country, it is computed as the arithmetic average of the values from the Government Effectiveness and Control of Corruption metrics divided by 100.

#### *Non-State Actors Coefficient*

For each country, it is computed as the arithmetic average of the normalised values of Freedom of Expression and Belief and Associational & Organisational Rights metrics.

Following, all coefficients are rescaled using 0.25 as the lower bound. This limits the ability of the coefficient to disproportionately reduce the value of the scores generated by primary data to ensure countries can retain at least 25% of their original score.

This is achieved by first multiplying the original coefficient value by 0.75 (which is the result of subtracting 0.25 from the maximum value of 1). Then, we add 0.25 to the result of that calculation, which raises the minimum possible value of the coefficient to 0.25, while keeping the maximum value at 1.

Finally, the adjusted value for each pillar within the thematic area is obtained as the product of the normalised indicators obtained in the previous step and the corresponding coefficient.

### **Step 4: Calculating the Score by Pillar**

The next step is to calculate the score by pillar for each country as the arithmetic average of the thematic areas.

For every country, each pillar (Government Frameworks, Government Actions, and Non-State Actors) has 19 indicators (one for each thematic area) ranging from 0 to 100. Since the indicators are equally weighted, the score for each pillar is determined by the arithmetic average of its 19 indicators. The result of this calculation returns a score ranging from 0-100, which allows for direct comparisons amongst the pillars and thematic areas in each country.

### **Step 5: Calculating the Index Score**

After computing the pillar scores, the next step is to calculate the Index Score as the weighted average of the pillars. To do this, it is necessary to assign each pillar a weight

that reflects their relative value in accordance with the conceptual framework of the GIRAI.

These weights are:

- Frameworks: 40%
- Government actions: 40%
- Non-state actors: 20%

Additionally, the thematic area and the dimension scores can also be calculated.

### **Step 6: Calculating the Score by Thematic Area**

The score for each thematic area is determined by calculating the weighted average of each of the pillars (Government Frameworks, Government Actions, and Non-state Actors). Each pillar is assigned its specific weight, which is then multiplied by the respective indicator scores, which are then added together to generate the overall score for the thematic area.

### **Step 7: Calculating the Dimension Score**

Finally, the dimension score is calculated by taking the arithmetic average of the scores for the thematic areas within the particular dimension.

## **Data Collection Methodology**

### **Global Research Network**

A Global Research Network, composed of 11 Regional Research Hubs (RRH) positioned at leading AI and technology research organisations across the world, and 138 national level researchers, was established to form the core human resources infrastructure for data collection. Recruitment and coordination of country-level researchers was led by the RRH, who further supported the continuity of training and data quality assurance. A dedicated individual within each Regional Research Hub acted as a “team leader” for the country-level researchers they recruited to perform data collection within their region.



## Training and Capacity Building

RRH were supported by the core team of the GIRAI to hold extensive training workshops for all researchers on the GIRAI methodology, survey instruments, quality assurance, and review processes. These sessions were held prior to data collection, and were accompanied by key training material as the [Researcher's Handbook](#), which contains the definitions and justifications of all the conceptual components of the dimensions, thematic areas and indicators that comprise the Global Index on Responsible AI.

## Primary Data Survey Tool

Data collection was conducted via detailed surveys and questionnaires structured to support comprehensive analysis across the primary data indicators. In the questionnaires, researchers were asked to fill in only primary data indicators, through questions provided with closed-ended standardised scored answer options. Responses to the questionnaire required supporting evidence, which were reviewed by the team leader, a board of external reviewers and the core team. The entire questionnaire was completed on an online survey tool – [Survey Solutions](#) – developed by the World Bank.

## Limitations of the Methodological Approach

It is important to note that indices, by their nature, simplify complex phenomena in order to be able to provide a global review of a concept, or set of concepts. Although the Index has been built to measure national AI ecosystems in a multi-dimensional way, scores are designed to reduce complex phenomena into a set of numerical rankings, which may simplify critical nuances in the data, such as other cross-cutting areas that may be applicable to AI even though they make no specific mention of it. Such nuances can reveal important insights in the scoring, highlight exceptions or outliers in the data, and offer a more precise interpretation of the results, which is useful for determining the value of the Index's findings, especially on a country and regional level.

The GIRAI's dependency on publicly available data has highlighted significant disparities among countries, influenced by factors like information regulations, resource accessibility, online information availability, and the currency and comprehensiveness of data capturing specific entities' activities. This variation in data accessibility may have influenced scoring, which may skew the focus of results towards government transparency over implementation responsible AI practices.



The Global Index on Responsible AI measures the existence of frameworks related to the key concepts of responsible AI, but not their actual efficacy to protect and promote human rights. In addition, the Global Index on Responsible AI does not measure the human rights impact of the use of specific AI systems in different country contexts.

Acknowledging the shortcomings of the Index not only helps put into context what it does measure and what it doesn't, but also provides the foundation for understanding how to improve its scope, methodology, and accuracy in the future.

## **Annexure A: Sample questionnaire: Gender equality thematic area**

### **SURVEY IDENTIFICATION INFORMATION**

#### **QUESTIONNAIRE DESCRIPTION**

- Dimension: Human Rights and AI
  - Sub-dimension: Civil and Political Rights
  - Thematic Area: Gender Equality
- 

#### **INSTRUCTIONS**

Before starting your work on this questionnaire, please download and read the complete guidance for this specific thematic area by clicking on the link below. There you will find a clear definition of the concept, examples of the evidence allowed in each indicator, and guidance on how to find them.

---

#### **GENDER EQUALITY – FRAMEWORKS**

Frameworks are one of the key pillars into which the concept of Responsible AI was broken down to measure it through the GIRAI survey. In this study, a "Framework" comprises "relevant laws, regulations, policies, strategies, and/or guidelines" that prescribe how different actors should act towards a topic, how the technology should be handled to protect human rights, channels of obtaining redress, and penalties.

Please provide the requested information in the following questions regarding laws, regulations, policies, strategies, and/or guidelines related to Gender Equality and Artificial Intelligence that were in effect and/or approved during the study's timeframe (November 1, 2021 – November 1, 2023).

---

Question 1: Are there national/federal laws, regulations, policies, strategies, plans, and/or guidelines that specifically address Artificial Intelligence and Gender Equality?

- No, there aren't. *(If selected, proceed to Question 2)*
- Yes, there are. *(Continue to Question 3)*

Question 2: Please explain how you arrived at this answer.

*Please explain the research steps taken to reach this conclusion. Remember that the answer can't exceed 750 characters. (Continue to question 39)*

Question 3: What is the title of the document in its original language?

*If you have used this document in other thematic areas, please always provide exactly the same title.*

Question 4: What is the title of the document in English?

*Provide the title of the document translated into English. If you have used this document in other thematic areas, please always provide exactly the same title.*

Question 5: What is the URL or web address where the document can be found?

*If you have used this document in other thematic areas, please always provide exactly the same URL.*

Question 6: What is the type of the document?

- Law
- Policy
- Strategy
- Guideline
- International Law
- Other (Specify)

Question 7: Please specify the other type.

Question 8: Which is the date of approval of the document?

*Date of approval should be before (1-11-2023).*

Question 9: Briefly explain how the document addresses the thematic area.

*Remember that the answer can't exceed 750 characters.*

Question 10: Reference the section and pages number that speak to this thematic area.

Question 11: What is the nature of the document in terms of enforceability?

- Non-Binding
- Binding
- Legally-Enforceable

Question 12: According to the document, briefly explain how you arrived at this answer.  
*Remember that the answer can't exceed 750 characters.*

Question 13: Reference the section and page number that speak to enforceability.

Question 14: To which sectors is the document applicable? *(Select all that apply.)*

- All state sectors
- Only specific sector(s) inside the state (Some ministries, agencies, etc.)
- All non-state sectors
- Only specific sector(s) inside the non-state sector (Some industries, fields, etc.)

Question 15: Reference the section and page number that speak to the sectors selected in the previous question.

Question 16: Does the document specifically mention any stages of the AI lifecycle?  
*(Select all that apply)*

- Design and development
- Adoption and use
- Implementation and impact
- Not mentioned in the document

Question 17: According to the document, briefly explain how you arrived at this answer.  
*Remember that the answer can't exceed 750 characters.*

Question 18: Reference the section and pages number that speak to the stages of the AI lifecycle selected in the previous question.

Question 19: Is there another document that covers any of the remaining sectors? *(only answer if in Question 14 one of the State or Non-state sector is not selected)*

- No, there isn't. *(Continue to Question 20)*
- Yes, there is. *(Continue to Question 21)*

Question 20: Please explain how you arrived at this answer.

*Please explain the research steps taken to reach this conclusion. Remember that the answer can't exceed 750 characters. (Proceed to Question 37)*

Question 21: What is the title of the document in its original language?

*If you have used this document in other thematic areas, please always provide exactly the same title.*

Question 22: What is the title of the document in English?

*Provide the title of the document translated into English. If you have used this document in other thematic areas, please always provide exactly the same title.*

Question 23: What is the URL or web address where the document can be found?

*If you have used this document in other thematic areas, please always provide exactly the same URL.*

Question 24: What is the type of the document?

- Law
- Policy
- Strategy
- Guideline
- International Law
- Other (Specify)

Question 25: Please specify the other type.

Question 26: Which is the date of approval of the document?

*Date of approval should be before (1-11-2023).*

Question 27: Briefly explain how the document addresses the thematic area.

*Remember that the answer can't exceed 750 characters.*

Question 28: Reference the section and pages number that speak to this thematic area.

Question 29: What is the nature of the document in terms of enforceability?

- Non-Binding
- Binding
- Legally-Enforceable

Question 30: According to the document, briefly explain how you arrived at this answer.  
*Remember that the answer can't exceed 750 characters.*

Question 31: Reference the section and page number that speak to enforceability.

Question 32: To which sectors is the document applicable? *(Select all that apply.)*

- All state sectors
- Only specific sector(s) inside the state (Some ministries, agencies, etc.)
- All non-state sectors
- Only specific sector(s) inside the non-state sector (Some industries, fields, etc.)

Question 33: Reference the section and pages number that speak to the sectors selected in the previous question.

Question 34: Does the document specifically mention any stages of the AI lifecycle?  
*(Select all that apply)*

- Design and development
- Adoption and use
- Implementation and impact
- Not mentioned in the document *(If selected continue to question 35)*

Question 35: According to the document, briefly explain how you arrived at this answer.  
*Remember that the answer can't exceed 750 characters.*

Question 36: Reference the section and pages number that speak to the stages of the AI lifecycle selected in the previous question.

Question 37: Please provide any additional information, context, or observations related to the answers provided in the previous section that could help better understand the responses. If interviews were necessary, please provide details here.  
*Please remember that the answer can't exceed 750 characters.*

---

## GENDER EQUALITY – GOVERNMENT ACTIONS

Government actions indicators assess government initiatives with the object of disseminating or enforcing rights or obligations related to the indicator and the specific thematic area assessed.

Please provide the required information in the following questions regarding actions taken by the national/federal government to address Artificial Intelligence and Gender Equality. Answers should only consider examples that have either occurred or were current during the timeframe of the study (November 1, 2021 – November 1, 2023).

---

Question 38: Is there evidence of actions of the national/federal government to address Artificial Intelligence and Gender Equality?

- No, there isn't. *(If selected, proceed to Question 39)*
- Yes, there is. *(Continue to Question 40)*

Question 39: Please explain how you arrived at this answer.

*Please explain the research steps taken to reach this conclusion. Remember that the answer can't exceed 750 characters. (Proceed to Question 61)*

Question 40: What is the title of the action in English?

*Provide the name of the action translated in English. If you have used this action in other thematic areas, please always provide exactly the same name.*

Question 41: What is the URL or web address where the evidence of the action can be found?

*If you have used this action in other thematic areas, please always provide exactly the same URL. In case it's a draft framework, please load the URL to the document.*

Question 42: What is the type of the action?

- Draft frameworks (encompasses drafts of laws, policies, strategies, and other related documents) (If selected, proceed to Question 46)
- Operational/annual performance plans
- Plans/programmes
- Projects
- Grants/subsidies
- Budget allocation
- Requests for proposals
- Research/reports
- Training / courses
- Establishing partnerships and collaborations
- Establishing commissions/expert working groups
- Parliament discussions
- Assigning the development of AI policy to a specific government body
- Establishment of government bodies
- Hearings
- Conferences / workshops
- Other (Specify)

Question 43: Please specify the other type of action.

Question 44: Geographical coverage of the action.

- National
- Sub-National
- Local

Question 45: Briefly explain how the action addresses the thematic area.

*Remember that the answer can't exceed 750 characters. (Continue to Question 61)*

Question 46: What is the title of the draft in its original language?

*If you have used this document in other thematic areas, please always provide exactly the same title.*

Question 47: What is the title of the draft in English?

*Provide the title of the document translated in English. If you have used this document in other thematic areas, please always provide exactly the same title.*



Question 48: What is the type of the draft?

- Law
- Policy
- Strategy
- Guideline
- International Law
- Other (Specify)

Question 49: Please specify the other type.

Question 50: Which is the date of publication of the draft?

*Date of the release or publication should be between (01-11-2021, 01-11-2023).*

Question 51: Briefly explain how the draft addresses the thematic area.

*Remember that the answer can't exceed 750 characters.*

Question 52: Reference the section and pages number that speak to this thematic area.

Question 53: If this draft document is finally approved, which is meant to be its nature in terms of enforceability?

- Non-Binding
- Binding
- Legally-Enforceable

Question 54: According to the draft, briefly explain how you arrived at this answer.

*Remember that the answer can't exceed 750 characters.*

Question 55: Reference the section and page number that speak to enforceability.

Question 56: To which sectors is the draft applicable? (*Select all that apply.*)

- All state sectors
- Only specific sector(s) inside the state (Some ministries, agencies, etc.)
- All non-state sectors
- Only specific sector(s) inside the non-state sector (Some industries, fields, etc.)

Question 57: Reference the section and pages number that speak to the sectors selected in the previous question.

Question 58: Does the draft specifically mention any stages of the AI lifecycle? (*Select all that apply*)

- Design and development
- Adoption and use
- Implementation and impact
- Not mentioned in the document

Question 59: According to the document, briefly explain how you arrived at this answer. Remember that the answer can't exceed 750 characters.

Question 60: Reference the section and pages number that speak to the stages of the AI lifecycle selected in the previous question.

Question 61: Please provide any additional information, context, or observations related to the answers provided in the previous section that could help better understand the responses. If interviews were necessary, please provide details here.

*Please remember that the answer can't exceed 750 characters.*

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## GENDER EQUALITY – NON-STATE ACTORS

Non-state actors indicators assess whether there are other actors than the government on the field working on the topic evaluated, such as research centers and think tanks, civil society organizations, advocacy groups and non-profits focusing on the ethical and social implications of AI, private sector companies, or professional associations gathering and sharing best practices and standards among its members, media, and journalists disseminating information and raising awareness about AI and its impact on society.

Please provide the requested information in the following questions about Non-state actors working towards AI and Gender Equality in the country. You can load one actor per each type, and for each actor one initiative must be provided. Please only include relevant actors who were active within the study's timeframe (November 1, 2021 – November 1, 2023) and focus on initiatives that occurred during this specific period.

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## Private Sector Actors

Question 62: Are there any actors from the private sector working towards Gender Equality and AI in the country?

- No, there aren't. *(If selected, proceed to Question 63)*
- Yes, there are. *(Continue to Question 64)*

Question 63: Please explain how you arrived at this answer.

*Please explain the research steps taken to reach this conclusion. Remember that the answer can't exceed 750 characters. (Proceed to Question 74)*

Question 64: What is the name of the actor in its original language?

*If you have mentioned this actor in other thematic areas, please always provide exactly the same name.*

Question 65: What is the name of the actor in English?

*If you have mentioned this actor in other thematic areas, please always provide exactly the same name.*

Question 66: Please provide the website of the actor.

*If you have mentioned this actor in other thematic areas, please always provide exactly the same URL.*

Question 67: What is the type of this private sector actor?

- State owned enterprise
- Company
- Other (Specify)

Question 68: Please specify the other type.

Question 69: What is the name of the initiative in English?

*If you have mentioned this initiative in other thematic areas, please always provide exactly the same name.*

Question 70: What is the URL or web address where evidence about this initiative can be found?

*If you have mentioned this initiative in other thematic areas, please always provide exactly the same URL.*

Question 71: What is the type of this initiative?

- Research and analysis
- Advocacy and public awareness campaigns
- Developing ethical guidelines, standards, and best practices
- Establishing partnerships and collaborations
- Legal assistance / action
- Non-state mechanisms
- Training / courses
- Conferences / workshops
- Other (Specify)

Question 72: Specify the other type of initiative.

Question 73: Briefly explain how the initiative addresses the thematic area.

*Remember that the answer can't exceed 750 characters.*

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Civil Society Actors

Question 74: Are there any actors from the civil society working towards Gender Equality and AI in the country?

- No, there aren't. *(If selected, proceed to Question 75)*
- Yes, there are. *(Continue to Question 76)*

Question 75: Please explain how you arrived at this answer.

*Please explain the research steps taken to reach this conclusion. Remember that the answer can't exceed 750 characters. (Proceed to Question 86)*

Question 76: What is the name of the actor in its original language?

*If you have mentioned this actor in other thematic areas, please always provide exactly the same name.*

Question 77: What is the name of the actor in English?

*If you have mentioned this actor in other thematic areas, please always provide exactly the same name.*

Question 78: Please provide the website of the actor.

*If you have mentioned this actor in other thematic areas, please always provide exactly the same URL.*

Question 79: What is the type of this civil society actor?

- Professional association
- Sectoral organisation
- Independent research institute
- Community-based organisation
- Faith-based organisation
- Social movement
- Labour union
- Other (Specify)

Question 80: Please specify the other type.

Question 81: What is the name of the initiative in English?

*If you have mentioned this initiative in other thematic areas, please always provide exactly the same name.*

Question 82: What is the URL or web address where evidence about this initiative can be found?

*If you have mentioned this initiative in other thematic areas, please always provide exactly the same URL.*

Question 83: What is the type of this initiative?

- Research and analysis
- Advocacy and public awareness campaigns
- Developing ethical guidelines, standards, and best practices
- Establishing partnerships and collaborations
- Legal assistance / action
- Non-state mechanisms
- Training / courses
- Conferences / workshops
- Other (Specify)

Question 84: Specify the other type of initiative.

Question 85: Briefly explain how the initiative addresses the thematic area.

*Remember that the answer can't exceed 750 characters.*

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#### Academia/University Actors

Question 86: Are there any actors from academia or university working towards Gender Equality and AI in the country?

- No, there aren't. *(If selected, proceed to Question 87)*
- Yes, there are. *(Continue to Question 88)*

Question 87: Please explain how you arrived at this answer.

*Please explain the research steps taken to reach this conclusion. Remember that the answer can't exceed 750 characters. (Proceed to Question 98)*

Question 88: What is the name of the actor in its original language?

*If you have mentioned this actor in other thematic areas, please always provide exactly the same name.*

Question 89: What is the name of the actor in English?

*If you have mentioned this actor in other thematic areas, please always provide exactly the same name.*

Question 90: Please provide the website of the actor.

*If you have mentioned this actor in other thematic areas, please always provide exactly the same URL.*

Question 91: What is the type of this actor?

- Public University
- Private University
- Research Council
- Research Institute
- Other (Specify)

Question 92: Please specify the other type.

Question 93: What is the name of the initiative in English?

*If you have mentioned this initiative in other thematic areas, please always provide exactly the same name.*

Question 94: What is the URL or web address where evidence about this initiative can be found?

*If you have mentioned this initiative in other thematic areas, please always provide exactly the same URL.*

Question 95: What is the type of this initiative?

- Research and analysis
- Advocacy and public awareness campaigns
- Developing ethical guidelines, standards, and best practices
- Establishing partnerships and collaborations
- Legal assistance / action
- Non-state mechanisms
- Training / courses
- Conferences / workshops
- Other (Specify)

Question 96: Specify the other type of initiative.

Question 97: Briefly explain how the initiative addresses the thematic area.

*Remember that the answer can't exceed 750 characters.*

Question 98: If there is any further information, context, or observations related to the responses given in the Non-State Actors section, please provide them here. If any interviews were conducted, please provide the relevant details.

*Please remember that the answer can't exceed 750 characters.*

## Annexure A: Worked example of Scoring Calculation

Let's take a country that does not have a National AI Policy but one is in development,, and where organisations from the civil society and universities are supporting this effort.

The coefficients for the country are as follows: Government Frameworks coefficient: 0.73, Government Actions coefficient: 0.76, Non-state actors coefficient: 0.98.

Let's calculate the scores following the steps previously detailed:

|                                                                                                                                                                    | Government Frameworks                                             | Government Actions                                                                                                                                                                                                          | Non-state Actors                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Step 1</b><br>convert qualitative data into numerical values<br><br><u>Explanation:</u> points are assigned according to the scoring rules                      | 0 points<br><br>As the country doesn't have a framework in place. | 1 point<br><br>Full point is given as the country doesn't have a framework in place, but is developing one. Note that if the action carried out by the government was of another type, the country would score 0.90 points. | 2 points<br><br>One point is awarded for each actor from a different sector. As in the example there are actors from two sectors, the country receives 2 points out of 3. |
| <b>Step 2</b><br>normalise raw scores into a 0-100 scale<br><br><u>Explanation:</u> points are normalised using a min - max formula                                | 0 points<br><br>$= (0-0) / (2-0) * 100$                           | 100.00 points<br><br>$= (1-0) / (1-0) * 100$                                                                                                                                                                                | 66.67 points<br><br>$= (2-0) / (3-0) * 100$                                                                                                                               |
| <b>Step 3</b><br>adjust the normalised scores using the pillar coefficients<br><br><u>Explanation:</u> points are adjusted by the corresponding pillar coefficient | 0 points<br><br>$= 0 * 0.73$                                      | 76.00 points<br><br>$= 100 * 0.76$                                                                                                                                                                                          | 65.34 points<br><br>$= 66.67 * 0.98$                                                                                                                                      |
| <b>SCORE</b>                                                                                                                                                       | 0 points                                                          | 76.00 points                                                                                                                                                                                                                | 65.34 points                                                                                                                                                              |



Now let's suppose we have the pillar scores for the 19 thematic. We could obtain the Pillars score for the country as follows.:

Step 4: calculate the Pillar scores as the arithmetic mean of the Thematic Area score within the pillar.

| Dimension                     | Thematic Area                           | Government Frameworks                          | Government Actions                          | Non-state Actors                         |
|-------------------------------|-----------------------------------------|------------------------------------------------|---------------------------------------------|------------------------------------------|
| 1) Responsible AI Governance  | 1) National AI Policy                   | 0.00                                           | 76.00                                       | 65.34                                    |
|                               | 2) Impact Assessments                   | 64.00                                          | 2.00                                        | 29.00                                    |
|                               | 3) Human Oversight and Determination    | 15.00                                          | 74.00                                       | 74.00                                    |
|                               | 4) Responsibility and Accountability    | 53.00                                          | 42.00                                       | 47.00                                    |
|                               | 5) Proportionality and Do No Harm       | 41.00                                          | 23.00                                       | 84.00                                    |
|                               | 6) Public Procurement                   | 9.00                                           | 100.00                                      | 59.00                                    |
|                               | 7) Transparency and Explainability      | 4.00                                           | 62.00                                       | 42.00                                    |
|                               | 8) Access to Remedy and Redress         | 22.00                                          | 36.00                                       | 60.00                                    |
|                               | 9) Safety, Accuracy and Reliability     | 19.00                                          | 51.00                                       | 36.00                                    |
| 2) Human Rights and AI        | 10) Gender Equality                     | 99.00                                          | 23.00                                       | 61.00                                    |
|                               | 11) Data Protection and Privacy         | 50.00                                          | 94.00                                       | 56.00                                    |
|                               | 12) Public Participation and Awareness  | 69.00                                          | 66.00                                       | 37.00                                    |
|                               | 13) Bias and Unfair Discrimination      | 82.00                                          | 89.00                                       | 34.00                                    |
|                               | 14) Children's Rights                   | 56.00                                          | 66.00                                       | 10.00                                    |
|                               | 15) Labour Protection and Right to Work | 49.00                                          | 61.00                                       | 1.00                                     |
|                               | 16) Cultural and Linguistic Diversity   | 80.00                                          | 21.00                                       | 41.00                                    |
| 3) Responsible AI Capacities  | 17) Competitions Authorities            | 2.00                                           | 97.00                                       | 45.00                                    |
|                               | 18) Public Sector Skills Development    | 24.00                                          | 5.00                                        | 12.00                                    |
|                               | 19) International Cooperation           | 80.00                                          | 34.00                                       | 37.00                                    |
| Average Thematic Areas 1 – 19 |                                         | 43.05<br>Government Frameworks<br>Pillar Score | 53.79<br>Government Actions<br>Pillar Score | 43.7<br>Non-state Actors<br>Pillar Score |

Step 5: calculate the Index score as the weighted mean of the Pillars.

$$\text{Index score} = (43.05 \text{ points} * 40\%) + (53.79 \text{ points} * 40\%) + (43.70 \text{ points} * 20\%)$$

$$\text{Index score} = 47.48 \text{ points}$$

As mentioned, it is also possible to calculate the Thematic Area and the Dimension score.

Step 6: The Thematic Area score is calculated as the weighted mean of the Pillars score at the Thematic Area level. For the National AI Policy Thematic Area this would be:

$$\text{National AI Policy score} = (0 * 40\%) + (76.00 \text{ points} * 40\%) + (65.34 \text{ points} * 20\%)$$

$$\text{National AI Policy score} = 43.47 \text{ points}$$

Step 7: The Dimension score is calculated as the arithmetic mean of the Thematic Areas that make each Dimension.

| Dimension                                                                | Thematic Area                           | Thematic Area Score |
|--------------------------------------------------------------------------|-----------------------------------------|---------------------|
| 1) Responsible AI Governance                                             | 1) National AI Policy                   | 43.47               |
|                                                                          | 2) Impact Assessments                   | 32.20               |
|                                                                          | 3) Human Oversight and Determination    | 50.40               |
|                                                                          | 4) Responsibility and Accountability    | 47.40               |
|                                                                          | 5) Proportionality and Do No Harm       | 42.40               |
|                                                                          | 6) Public Procurement                   | 55.40               |
|                                                                          | 7) Transparency and Explainability      | 34.80               |
|                                                                          | 8) Access to Remedy and Redress         | 35.20               |
|                                                                          | 9) Safety, Accuracy and Reliability     | 35.20               |
| Responsible AI Governance Score (average Thematic Areas 1 – 9)           |                                         | 41.83               |
| 2) Human Rights and AI                                                   | 10) Gender Equality                     | 61.00               |
|                                                                          | 11) Data Protection and Privacy         | 68.80               |
|                                                                          | 12) Public Participation and Awareness  | 61.40               |
|                                                                          | 13) Bias and Unfair Discrimination      | 75.20               |
|                                                                          | 14) Children's Rights                   | 50.80               |
|                                                                          | 15) Labour Protection and Right to Work | 44.20               |
|                                                                          | 16) Cultural and Linguistic Diversity   | 48.60               |
| Human Rights and AI Score (average Thematic Areas 10 –16)                |                                         | 58.57               |
| 3) Responsible AI Capacities                                             | 17) Competitions Authorities            | 48.60               |
|                                                                          | 18) Public Sector Skills Development    | 14.00               |
|                                                                          | 19) International Cooperation           | 53.00               |
| National Responsible AI Capacities Score (average Thematic Areas 17 –19) |                                         | 38.53               |